

Assignment

1. What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?

Answer:

Native, Hybrid, and Cross-platform are different ways of developing mobile apps.

- **Native Development** means creating an app specifically for one operating system, such as Android or iOS. It provides the best performance, smooth user experience, and full access to device features. It is commonly used for games, banking apps, and applications that need high performance.
- **Hybrid Development** uses web technologies like HTML, CSS, and JavaScript, then wraps them into a mobile app. One codebase works on different platforms, making development faster and cheaper. It is commonly used for simple business apps, prototypes, and apps with basic features.
- **Cross-platform Development** uses frameworks like Flutter or React Native to build apps for Android and iOS from a single codebase. It offers better performance than hybrid apps while reducing development time. It is commonly used for startups, business applications, and projects that need to support multiple platforms with one codebase.

2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?

Answer:

The Flutter Ecosystem is the collection of tools, libraries, packages, and services that support Flutter app development. It includes the Flutter SDK, Dart programming language, widgets, plugins, and packages available on pub.dev.

Flutter helps developers create modern mobile applications by allowing them to write one code that works on Android, iOS, web, and desktop. It also provides ready-made widgets for attractive user interfaces, fast development through Hot Reload, and good performance because Flutter compiles directly into native code.

3. What is the Software Development Life Cycle (SDLC), and why is it important in building successful applications?

Answer:

The Software Development Life Cycle (SDLC) is a step-by-step process used to develop software efficiently and with high quality.

The main stages of SDLC are:

1. Planning

2. Requirements Analysis
3. Design
4. Development
5. Testing
6. Deployment
7. Maintenance

SDLC is important because it helps developers organize their work, reduce mistakes, improve software quality, manage time effectively, and make sure the final application meets the users' needs.

4. How do Flutter development practices and the SDLC work together in creating real-world mobile applications?

Answer:

Flutter development practices and the SDLC work together because Flutter is the tool used to build the application, while the SDLC provides the process for developing it correctly.

During the planning and design stages, developers decide what features the app will have. During development, Flutter is used to build the application's interface and functionality. Testing ensures the app works properly on different devices, and after deployment, Flutter makes it easier to update and maintain the application.

By combining Flutter with the SDLC, developers can create applications that are well-planned, reliable, and easier to improve over time.

5. Why should developers understand these concepts before starting an application project?

Answer:

Developers should understand these concepts before starting a project because they help them choose the right development approach, use the best tools, and follow an organized process. Knowing the differences between Native, Hybrid, and Cross-platform development helps in selecting the most suitable technology for the project. Understanding the Flutter Ecosystem allows developers to build applications more efficiently, while learning the SDLC helps ensure that the project is completed successfully with fewer errors. Overall, these concepts improve the quality of the application and increase the chances of building a successful project.